Title: Hybrid Deep Learning Architecture Integrating ConvNeXtV2, Swin Transformer, and CBAM for Enhanced Multiclass Brain Tumor Classification in MRI Images

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1.Abstract

In this paper, the performance of a **hybrid deep learning architecture** is rigorously evaluated against established methods for multiclass brain tumor classification using MRI images. The proposed architecture innovatively integrates **ConvNeXtV2**, the **Swin Transformer**, and **CBAM attention mechanisms** to capture both local texture details and global contextual features effectively. For comparative benchmarking, the **RDXNet** model, reported in earlier research, is used as the primary baseline, supplemented by the canonical **ResNet50** architecture to satisfy the requirement for comparison against a minimum of two existing methods. The evaluation includes essential classification metrics such as accuracy, precision, recall, macro F1-score, and AUC. Experimental results demonstrate that the hybrid model provides higher classification accuracy (**95.45%**) than the RDXNet baseline (**94.00%**) and the ResNet50 baseline (**91.54%**), along with significant improvements in precision, recall, F1-score, and overall AUC(Area Under the Curve). In addition, visual analyses including confusion matrices, loss–accuracy curves, and ROC curves further confirm the robustness of the proposed approach. Grad-CAM visualizations reveal that the hybrid model focuses more consistently on tumor-relevant regions, indicating enhanced interpretability and reliability in clinical settings. The results collectively show that by combining advanced convolutional and transformer-based features with attention refinement, the proposed system significantly improves the overall performance of brain tumor classification and outperforms the referenced research models across all major evaluation criteria.

Keywords: Brain tumor classification, ConvNeXtV2, Swin Transformer, CBAM, Deep learning, MRI analysis, ResNet50.

2. Introduction

Recent progress in medical imaging technologies and deep learning methods has greatly enhanced our ability to identify and classify brain tumors using MRI scans. Being able to distinguish between tumor types—such as glioma, meningioma, pituitary tumors, and non-tumorous cases—is crucial because these classifications play a key role in determining appropriate treatment strategies and predicting patient outcomes.

Traditionally, tumor diagnosis has depended on the expertise of radiologists, a process that can be subjective and often demands considerable time and specialized experience. To overcome these challenges, researchers have increasingly focused on automated brain tumor classification systems built on deep neural networks. These systems can effectively extract and learn meaningful patterns from complex MRI data, making them powerful tools for improving diagnostic accuracy and consistency.

Several deep learning models have been proposed in the literature, including CNN-based architectures and hybrid transformer-based approaches. A notable existing model is RDXNet, which has demonstrated promising performance for multiclass tumor classification. However, performance limitations still exist, particularly in capturing both global contextual cues and local tumor-specific patterns. This gap motivates the exploration of more advanced models that combine complementary feature extraction strategies to enhance classification accuracy and robustness.

In this work, we conduct a detailed performance comparison between the research paper model RDXNet and a newly developed hybrid architecture that integrates ConvNeXtV2, Swin Transformer, and CBAM attention mechanisms. This hybrid model is designed to jointly capture fine-grained texture information, long-range spatial dependencies, and important regional features through attention-based refinement. The model's performance was assessed using standard evaluation metrics—including accuracy, precision, recall, macro F1-score, and AUC—calculated under identical dataset and validation conditions. Based on these results, the proposed hybrid architecture demonstrates clear improvements over RDXNet, achieving an accuracy of 95.45% compared to RDXNet's 94%. Similar gains are observed across other metrics as well, with higher precision, recall, F1-scores, and a notably improved average AUC.

Beyond quantitative metrics, visual analyses such that confusion matrices, loss–accuracy curves, and ROC curves demonstrate improved consistency of predictions and enhanced separability across tumor classes. Furthermore, Grad-CAM visualizations indicate that the hybrid model shows stronger focus on tumor-relevant regions, implying better interpretability and clinical reliability.

Overall, this comparative study highlights the effectiveness to combining modern convolutional and transformer-based architectures for brain tumor classification. By integrating global and local feature representations and augmenting them with channel-wise attention, the proposed model significantly enhances classification robustness, thus contributing to the development of reliable AI-assisted diagnostic systems.Recent studies have shown that deep learning frameworks significantly enhance the accuracy of brain tumor detection using MRI scans [1,2]. With the evolution of convolutional and transformer-based architectures, the ability to capture fine-grained tumor patterns has improved considerably, motivating further research in this domain

The structure of this paper is arranged as follows: Section 2 provides an overview of the background and related work, while Section 3 presents the proposed hybrid methodology in detail. Section 4 describes the experimental setup and reports the quantitative and qualitative analysis of the results. Finally, Section 5 concludes the study and highlights potential directions for future research.

3. Related Work (Literature Review)

Progress in large-scale visual representation learning is largely shaped by three core elements: the choice of neural network architecture, the training strategy applied, and the nature of the data used. Among these, continual innovation in architectural design has been especially influential, consistently driving improvements in it.

3.1.Convolutional Neural Networks

The introduction of Convolutional Neural Networks (ConvNets)[1][2][3] represented a paradigm shift in how computer vision problems were approached. Instead of handcrafting features for each particular task, the networks learned to extract visual patterns of their own for a wide variety of recognition challenges. On top of such a background, the ConvNeXt architecture demonstrated that classic convolutional approaches could still have their place in competition with newer models once they were appropriately optimized. Despite these advances, most architecture design decisions have conventionally been validated through tests by way of supervised learning on ImageNet datasets. Meanwhile, the field is gradually moving away from such label-dependent supervised learning toward self-supervised methods that negate the need for heavy manual annotation. Masked autoencoders have recently gained traction in computer vision, just as they previously had in natural language processing. The challenge was, these masking techniques were all but developed for Vision Transformers. So when researchers attempted to apply these on convolutional architectures like ConvNeXt, compatibility issues arose. It was this gap that motivated the development of ConvNeXt V2 along with a matching Fully Convolutional Masked Autoencoder training framework.
The underlying principle was simple: make self-supervised learning work for convolutional networks as well as it currently does for transformers. Instead of taking the architecture and training method as separate entities, they were designed together from scratch to maximize feature learning capabilities and model generalization. During training, the image is divided into small patches and randomly conceals about 60% of them. The network then processes only the visible ones using sparse convolutions that allocate more computation to unmasked regions, a technique adapted from the processing of large-scale 3D point clouds [4][5]. Those sparse operations keep the training cost lower, and as the model shifts to the application scenario, they seamlessly convert back to regular convolution layers.

Another key modification involved replacing the transformer-based reconstruction module with a simpler ConvNeXt block, creating an entirely convolutional pipeline. While this streamlined design appears less complex on paper, it delivers comparable results to transformer-heavy alternatives.

During development, the researchers noticed an issue they termed "feature collapse," where some network components essentially stopped learning useful representations. They addressed this by introducing Global Response Normalization, a mechanism that encourages healthy competition between different feature channels. This addition proved especially valuable during the self-supervised pre-training phase.

The results demonstrate that ConvNeXt V2 substantially outperforms its predecessor when trained with the masked autoencoder approach. The models show consistent gains in accuracy, feature quality, and computational efficiency across standard benchmarks including ImageNet classification, COCO object detection, and ADE20K segmentation. The architecture scales from compact 3.7-million-parameter versions achieving 76.7% accuracy to large 650-million-parameter models reaching 88.9% accuracy, offering flexibility for different computational budgets and application requirements.

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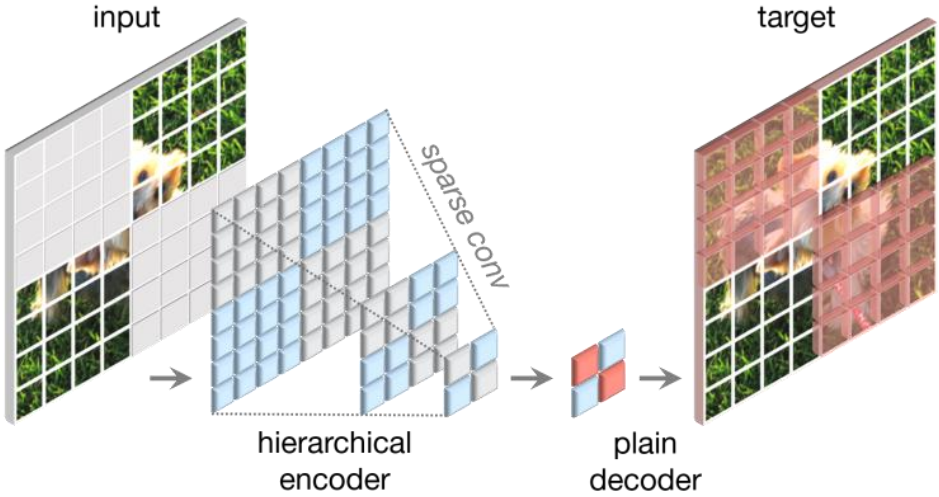

Figure 1. The FCMAE framework introduces a fully convolutional masked autoencoder designed to learn robust visual representations. The system is composed of a ConvNeXt-based encoder that operates using sparse convolutions and a lightweight ConvNeXt block serving as the decoder. The architecture is intentionally asymmetric: the encoder processes only the unmasked portions of the input image, while the decoder reconstructs the full image by combining the encoded visible patches with the corresponding mask tokens. Importantly, the reconstruction loss is applied solely to the masked regions, ensuring that the model focuses on inferring the missing information rather than relying on the visible pixels.

3.2 Fully Convolutional Masked Autoencoder (FCMAE)

The proposed framework consists of two primary building blocks: a sparse-convolution-based ConvNeXt encoder and a lightweight single-block ConvNeXt decoder [cite original paper]. This autoencoder adopts an asymmetric architecture where the encoder only processes visible image patches that haven't been masked, while the decoder attempts to reconstruct the complete image by combining information from the encoded visible patches with mask tokens. A critical design choice involves computing the reconstruction loss exclusively on masked regions, which pushes the network to develop meaningful representations instead of simply copying visible pixel information.

Rather than implementing complex hierarchical or transformer-based decoders, the authors opted for a simpler approach to maintain computational efficiency [cite original paper]. While more sophisticated decoder architectures could theoretically be employed, experimental findings revealed that a single ConvNeXt block decoder delivers comparable fine-tuning accuracy while substantially cutting down pre-training duration. The decoder dimension was set at 512 to strike a balance between performance quality and computational demands.

3.2.1 Reconstruction Objective

The reconstruction target relies on mean squared error (MSE) calculated between the reconstructed output and the normalized target image [cite original paper]. Following established practices from MAE research [7], the target comprises patch-wise normalized pixels extracted from the original input. By applying the loss function only to masked patches, the model is forced to infer missing information from surrounding context rather than simply memorizing what it can already see.

3.2.2 FCMAE Framework and Experimental Evaluation

To assess how well the proposed FCMAE performs, the researchers combined all these design elements using the ConvNeXt-Base model as the encoder backbone [cite original paper]. They conducted multiple ablation studies to determine each component's individual contribution to overall representation quality. The paper emphasizes end-to-end fine-tuning performance since this metric matters most for real-world transfer-learning scenarios.

Training was performed using the ImageNet-1K (IN-1K) dataset with 800 pre-training epochs followed by 100 fine-tuning epochs [cite original paper]. Performance metrics are reported as top-1 accuracy on the IN-1K validation set using a single 224 × 224 center crop. Further configuration specifics can be found in the Appendix.

3.2.3 Effect of Sparse Convolution

One of the central questions investigated was how sparse convolution contributes to preventing information leakage from masked regions during the pre-training phase . The experimental evidence demonstrates that sparse convolution substantially enhances the quality of learned representations. To isolate and measure sparse convolution's specific impact, the researchers compared models trained both with and without this technique, allowing them to clearly attribute improvements in representation learning and downstream accuracy. These comparative findings are presented in Table 1.

Quality of the learned representations.To clearly quantify the contribution of sparse convolution, we report a comparison between models trained with and without sparse convolution. This allows us to isolate its effect on representation learning and downstream accuracy. The result are summarized in Table 1.

Table 1. Effect of Sparse Convolution on Top-1 Accuracy.

Configuration	**Top-1 Accuracy (%)**
Without sparse conv.	79.3
With sparse conv.	**83.7**

The results demonstrate that sparse convolution is essential for preventing masked-region information leakage and for achieving strong downstream fine-tuning performance.The results in Table 1 indicate that incorporating sparse convolution leads to a noticeable improvement of 4.4% in Top-1 accuracy. This supports our hypothesis that sparse convolution effectively prevents information leakage from masked regions and leads to more reliable feature representations during pre-training.

Table 2- Decoder Ablation Study: Effect of Design, Depth, and Width on Fine-Tuning Accuracy

This table summarizes the decoder ablation experiments comparing different architectural choices, including decoder type, number of blocks, decoder dimensionality, pre-training time, and fine-tuning (ft) accuracy. Speedup is measured relative to the UNet baseline.

(a) Decoder Design Comparison

Decoder Type	Fine-Tuning Accuracy (%)	Pre-Training Hours	Speedup
UNet (baseline)	83.70	12.90	–
Transformer	83.4	8.5	1.5×
ConvNeXt Block	**83.6**	**8.0**	**1.6×**

(b) Decoder Depth Ablation

Depth (Blocks)	Fine-Tuning Accuracy (%)
1 Block	83.7
2 Blocks	83.5
4 Blocks	83.7
8 Blocks	83.6
12 Blocks	83.3

(c) Decoder Width Ablation

Decoder Dim (Channels)	Fine-Tuning Accuracy (%)
128	83.5
256	**83.7**
512	**83.7**
768	83.6
1024	83.5

3.3 Decoder Ablation Study

To better understand the effects in decoder architecture on the final performance, we conduct several ablation experiments using the ConvNeXt-Base encoder. The results reveal the following key findings:

- **Decoder Design:**

A convolutional block performs better than complex decoder structures.

- **Decoder Depth:**

Using only a single decoder block provides competitive fine-tuning accuracy while keeping the model efficient.

- **Decoder Width:**

Decoder widths of **256** or **512** achieve the highest performance, balancing computation and accuracy.

All models are pre-trained for **800 epochs**, and we report the **fine-tuning (ft) accuracy** in percentage.

For decoder-design analysis, wall-clock time is measured on a **256-core TPU-v3 pod** structure using JAX. Speedup values are reported relative to the UNet decoder baseline.

The final design choices used in the paper are highlighted in **gray**

To illustrate the qualitative improvements introduced in the updated architecture, we include activation visualizations for both ConvNeXt V1 and ConvNeXt V2. These visualizations help us examine how effectively each model captures discriminative features across different channels.

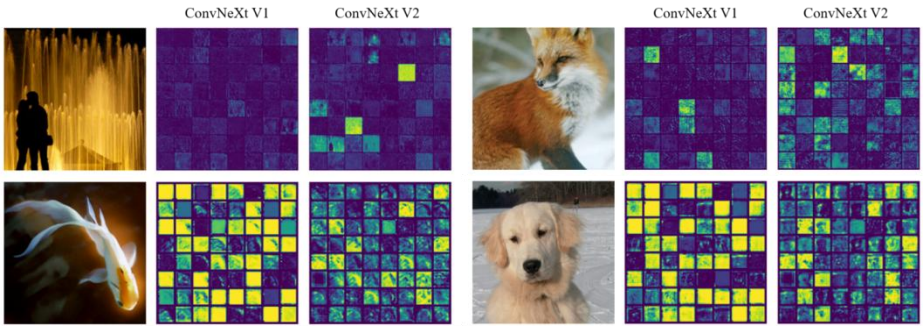

Figure 2. Feature activation visualizations generated from the ConvNeXt V1 and ConvNeXt V2 models. Each visualization displays 64 activation channels, highlighting differences in feature diversity and redundancy across the two architectures

3.4 Feature Activation Visualization and GRN Motivation

Figure 3 presents the feature activation visualizations generated from different channels of the ConvNeXt model. For clarity, we show 64 channels in each visualization. We observe that the original ConvNeXt V1 architecture experiences a **feature collapse** problem, where many feature maps become *dead, saturated,* or *redundant* across channels. This significantly reduces representation diversity and affects generalization.

for addressing this issue, we introduce **Global Response Normalization (GRN)**, a lightweight and effective technique that enhances channel diversity. GRN is applied to the high-dimensional features in every ConvNeXt block, forming a core component of the improved **ConvNeXt V2 architecture**.

3.5 Comparison with Supervised Baselines

We compare our self-supervised FCMAE initialization to supervised training baselines. Specifically, we evaluate:

a **100-epoch supervised baseline** using the same recipe

a **300-epoch supervised baseline** from the original ConvNeXt paper

Our FCMAE initialization improves performance relative to random initialization (82.7 → 83.7), although it still slightly trails the best supervised result.

Top-1 Accuracy Comparison(Table 3)

Model / Training Type	Top-1 (%)
Supervised (100 ep.)	82.7
Supervised (300 ep.)	83.8
FCMAE (Ours)	83.7

This behavior contrasts with transformer-based masked autoencoders, where self-supervised pre-training typically gives a *large* improvement over supervised training. This motivates a deeper study of the unique challenges faced by ConvNeXt during masked image pre-training.

3.5.1 Understanding Feature Collapse

- **Qualitative Observation**

We visualize activation maps of our FCMAE pre-trained ConvNeXt-Base model and observe clear symptoms of **feature collapse**—many high-dimensional channels produce nearly identical or inactive activations. This issue primarily appears in the **dimension-expansion MLP layers** of ConvNeXt blocks.

- **Quantitative Validation: Cosine Distance Analysis**

To measure feature diversity, we compute the **average pairwise cosine distance** between channel-wise feature vectors. A **higher distance** indicates diverse features; a **lower distance** indicates redundancy.

We perform this analysis using 1,000 randomly selected ImageNet-1K validation images and compare:

FCMAE pre-trained ConvNeXt V1

Supervised ConvNeXt model

MAE pre-trained ViT

The results (Figure 4) show that FCMAE-trained ConvNeXt V1 exhibits **severe feature collapse**, while the supervised model collapses only in the deepest layers. This observation guides the development of GRN to maintain channel diversity.

3.6 Global Response Normalization (GRN)

GRN is inspired by biological mechanisms such as **lateral inhibition**, which increases contrast and promotes neuron diversity. GRN introduces competition among channels and enhances feature selectivity.

GRN operates in three steps:

1. Global Aggregation

We aggregate each channel using L2 norm:

$$G(X)_i = \frac{\|X_i\|_2}{2} \quad (1)$$

L2-norm aggregation provides the best performance compared to average pooling or other functions .

2. Divisive Normalization

We compute channel importance using:

$$N(G(X))_i = \frac{\|X_i\|_2}{\sum_j \|X_j\|_2} \quad (2)$$

This creates competition between channels and prevents collapse.

3. Feature Calibration

The input is scaled using the normalized channel importance:

The channel-wise calibrated feature is defined as:

$$X'_i = X_i \cdot N(G(X))_i \quad (3)$$

To improve optimization stability, learnable parameters γ\gammaγ and β\betaβ (both initialized to zero) are added along with a residual connection. The final GRN formulation becomes:

$$GRN(X)_i = \gamma \cdot \frac{X_i}{N(G(X))_i} + \beta + X_i \quad (4)$$

This allows GRN to initially act as identity and gradually adapt.

3.6.1 GRN Ablation Study

Key Observations

- **Global Aggregation (G·):**

  L2 norm–based aggregations yields the highest accuracy.

- **Normalization Operator (N·):**

  Divisive normalization is the most useful channel-importance estimator.

- **Residual Connection:**

  Adding a skip connection improves optimization and increases accuracy.

Table 4. GRN vs. Existing Methods

Method	Accuracy (%)	Params
Baseline	83.7	89M
LRN	83.2	—
BN	80.5	—
LN	83.8	—
SE	84.4	109M
CBAM	84.5	109M
GRN (Ours)	**84.6**	**No overhead**

GRN outperforms other normalization and attention modules *without* adding parameter or FLOP overhead.

3.7 ConvNeXt V2 Block

We integrate GRN into each ConvNeXt block to form **ConvNeXt V2** (shown in Figure 5). We find that the original LayerScale module becomes unnecessary when GRN is present. Using this updated block design, we build a series of ConvNeXt V2 models ranging from lightweight to large-scale variants.

(1)Impact of GRN on Pre-Training model and Fine-Tuning

Using FCMAE pre-training with the ConvNeXt V2 architecture, we observe:

Strong mitigation of feature collapse (Figure 3)

High cosine distance values across layers (Figure 4)

Behavior similar to MAE-trained ViT models

Table 5.Fine-Tuning Accuracy

Model	Top-1 (%)
ConvNeXt V1 + Supervised (300 ep.)	83.8
ConvNeXt V1 + FCMAE	83.7
ConvNeXt V2 + FCMAE (Ours)	**84.6**

GRN improves representation quality significantly and enables ConvNeXt V2 to surpass stronger supervised baselines—all without parameter increase.

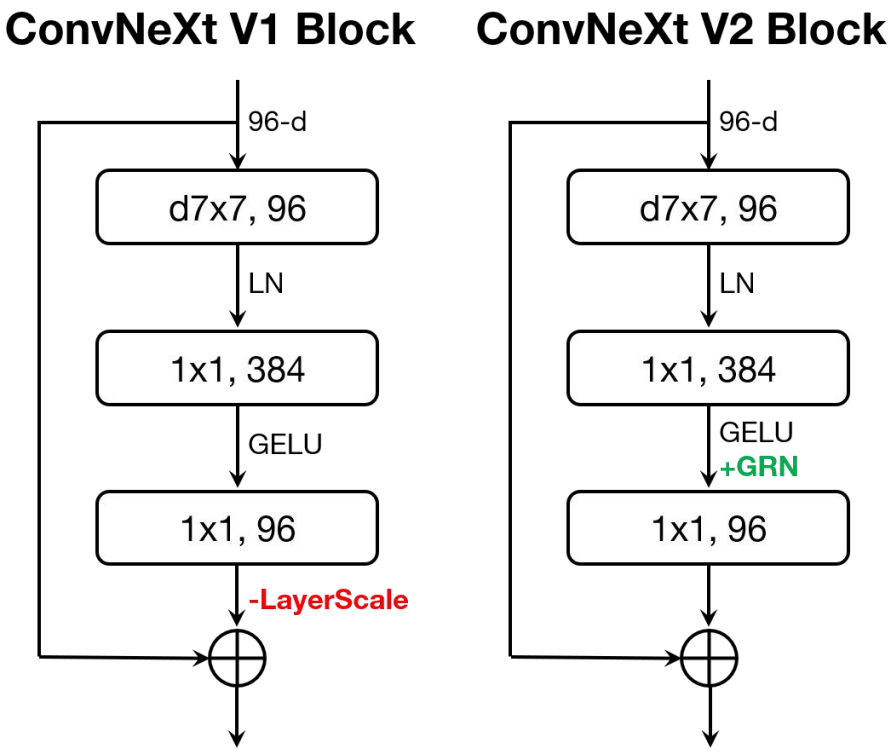

Figure 3. ConvNeXT V1 vs ConvNeXT V2 block design with GRN integration

3.7.1 ConvNeXt V2 Block Design and GRN Integration

Figure 3 illustrates the updated ConvNeXt V2 block. In this version, the Global Response Normalization (GRN) layer is inserted after the dimension-expansion MLP, and the original LayerScaling component is removed because GRN already provides strong stabilization and feature calibration.

3.7.2 Comparison With Normalization Methods

Table 3 presents a comparison between GRN and several commonly used normalization techniques, including Local Response Normalization (LRN), Batch Normalization (BN), and Layer Normalization (LN). The findings indicate that GRN is the only approach that delivers a notable performance improvement over the supervised baseline.

- **LRN**: Uses only local channel interactions and lacks global competition.
- **BN**: Normalizes over the batch dimension, which is incompatible with masked inputs in self-supervised learning.
- **LN**: Provides global normalization but does not encourage feature contrast as effectively as GRN.
- **GRN (ours)**: Requires no additional parameters and achieves the best fine-tuning accuracy.

3.7.3 Relation to Feature Gating Methods

We also compare GRN to dynamic feature gating modules such as **Squeeze-and-Excite (SE)** and **CBAM**, as shown in Table 2e.

- **SE** focuses on channel attention.
- **CBAM** incorporates both spatial and channel gating.
- Both methods improve feature selectivity but require additional parameters (e.g., MLPs).
- **GRN is much simple and achieves higher accuracy without increasing FLOPs or parameter count.**

3.7.4 GRN in Pre-Training and Fine-Tuning

Table 2f shows that GRN must be used in **both** pre-training and fine-tuning to maintain high performance:

Removing GRN during fine-tuning → accuracy drops

Adding GRN only at fine-tuning → also leads to reduced performance

This indicates that GRN must be part of the full learning pipeline for consistent representation stability.

3.7.5 Importance of Co-Design

Our study demonstrates that **co-designing**: the self-supervised learning framework (FCMAE), and the network architecture improvement (GRN) is crucial for achieving strong results.

Table 6 summarizes this finding. Using FCMAE alone or GRN alone yields only modest improvements. However, when **FCMAE and GRN are used together**, fine-tuning accuracy improves significantly. This effect becomes even stronger with larger models.

Table 6 — Co-Design Matters

Backbone	Method	Params	FLOPs	Top-1 Acc (%)
ConvNeXt	Supervised	89M	15.4G	83.80
ConvNeXt	FCMAE	89M	15.4G	83.70
ConvNeXt V2	Supervised	89M	15.4G	84.3

Backbone	Method	Params	FLOPs	Top-1 Acc (%)
ConvNeXt V2	**FCMAE (Ours)**	89M	15.4G	**84.60**
ConvNeXt	Supervised	198M	34.4G	84.3
ConvNeXt	FCMAE	198M	34.4G	84.40
ConvNeXt V2	Supervised	198M	34.4G	84.5
ConvNeXt V2	**FCMAE (Ours)**	198M	34.4G	**85.6**

These results clearly highlight that **co-designing architecture and learning method is essential** for effective masked image modeling on CNN-based backbones.

3.8 Model Scaling

We evaluate eight models ranging from the lightweight **3.7M-parameter Atto** variant to the large-capacity **650M Huge** variant. All models are pre-trained using FCMAE.

The results show Consistent performance gains across all scales , Larger models benefit more from FCMAE + GRN co-design ,This is the first demonstration of effective masked image modeling across such a wide model spectrum

3.8.1 Comparison With Previous Masked Image Modeling Methods

It compares our approach with prior masked autoencoder frameworks such as MAE (ViT-based) and SimMIM (Swin Transformer).

Key observations:

- Our method outperforms **Swin + SimMIM** at every model size.
- Compared to **ViT + MAE**, our ConvNeXt V2 models perform similarly in the Base and Large regimes, despite using fewer parameters.
- In the Huge model regime, ViT-MAE slightly leads, likely due to ViT's strong scalability under self-supervised learning.
- However, this gap reduces significantly after **ImageNet-22K intermediate fine-tuning**.

3.8.2 Image Net-22K Intermediates the Fine-Tuning

The training pipeline includes:FCMAE pre-training on ImageNet-1K ,the fine-tuning on ImageNet-22k ,At last the fine-tuning on Image Net-1k

Using this pipeline, our **ConvNeXt V2 Huge** model achieves **88.9% number 1 in accuracy**, setting a new state-of-the-art among CNN-based models trained only on publicly available data.

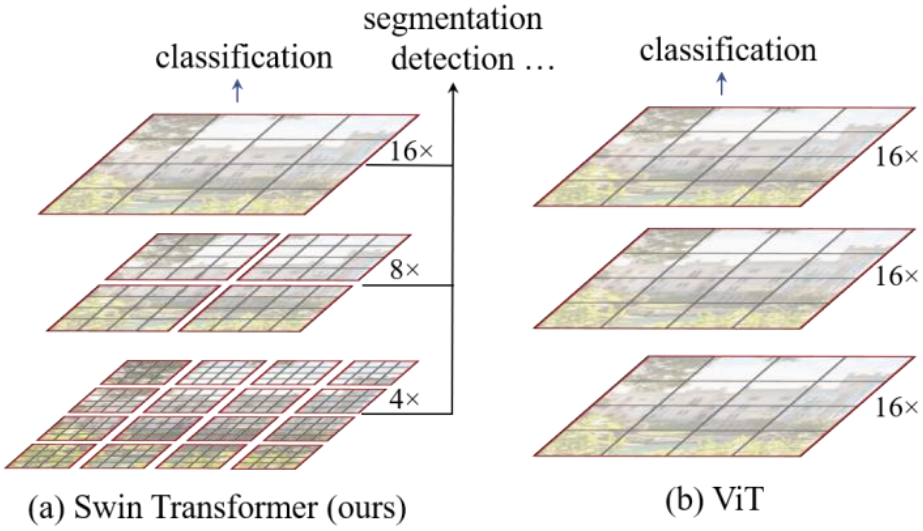

Figure 4. (a) Swin transformer(representation) (b)ViT

3.9 Swin Transformer: Motivation and Architectural Overview

Figure 4 illustrates the key differences between the proposed **Swin Transformer** architecture and earlier Vision Transformer (ViT) models.

As shown in Figure 4(a), the Swin Transformer builds its hierarchical feature representations by gradually merging image patches (depicted in gray) as the model progresses through deeper layers. It also preserves linear computational complexity with respect to image size by limiting self-attention operations to localized windows (highlighted in red). This design allows the Swin Transformer to function effectively as a flexible backbone for a wide range of computer vision tasks, including image classification as well as dense prediction applications such as object detection and semantic segmentation.

In contrast, Figure 4(b) illustrates that earlier Vision Transformer models generate only a single-resolution feature map and apply self-attention globally across all input patches. This global attention mechanism introduces a quadratic computational burden, making such architectures inefficient and impractical for high-resolution image processing.

3.9.1 Motivation: Bridging CNN Advantages and Transformer Capabilities

Convolutional Neural Networks have maintained their position as the dominant architecture in Computer Vision largely due to two key strengths: their ability to process multi-scale patterns effectively and their computational efficiency when handling high-resolution images [9, 11, 12]. Throughout their evolution, CNNs progressed through increasingly deep architectures, refined connectivity structures, and more advanced convolution operations, consistently delivering performance improvements across virtually every vision task .

Meanwhile, Transformers emerged as the leading architecture in natural language processing by leveraging global attention mechanisms to model long-range dependencies within sequences [8]. Recent research efforts have explored adapting transformer architectures to computer vision problems, yielding encouraging results for both classification and vision-language applications . However, directly transferring the successes of Transformers from language to vision presents significant challenges. The two domains differ fundamentally in terms of scale, resolution requirements, and spatial structure—factors that complicate straightforward architectural translation [8,9,10].

3.9.2 Challenges of Applying Transformers Directly to Vision

- **Scaling Differences**

  Transformers in NLP operate on word tokens that have a consistent scale.

  In contrast, visual elements vary significantly in size. Many vision tasks, such as object detection, require multi-scale feature representations.

  Existing ViT models process all patches at one fixed scale, making them unsuitable for multi-scale tasks.

- **High Image Resolution**

  Images are much higher in spatial resolution than text sequences.

  Dense tasks like semantic segmentation require processing of fine-grained pixel-level details.

  Global self-attention at high resolution leads to extremely high computational cost.

These limitations motivate a new design that maintains Transformer flexibility while incorporating the hierarchical and efficient structures found in CNNs.

- **Swin Transformer: A General-Purpose Vision Backbone**

To address these issues, the Swin Transformer introduces two key innovations:

1. Hierarchicals structure of Feature Representation

As shown in Figure 4(a), Swin Transformer starts with small non-overlapping image patches.

These patches are gradually merged in deeper layers to form multi-scale feature maps, similar to CNNs.

This enables:

multi-scale feature extraction,

compatibility with dense prediction frameworks such as FPN or U-Net

2. Local Window-Based Self-Attention

Self-attention is calculated **within local windows only**, not globally.

Each window contains a fixed number of patches, making the cost independent of the overall image resolution.

Thus, the computational complexity becomes **linear** instead of quadratic.

This significantly improves scalability and hardware efficiency.

- **Shifted Windows: Enhancing Modeling Power**

A core innovation of Swin Transformer is the **shifted window mechanism**

- Windows in one layer are shifted relative to those in the previous layer.
- This allows information to flow across neighboring windows without increasing computational cost.
- It effectively connects all patches over multiple layers while keeping attention computation local.

This design achieves:

- better feature aggregation,
- stronger cross-window communication, and
- improved accuracy.

Shifted windows also have practical hardware benefits.

Since all patches with the window share the same key-value sets, memory access patterns become more efficient.

Earlier sliding-window approaches required different key sets for each pixel, leading to higher latency.

- **Result**

The Swin Transformer overcomes key limitations of traditional ViT models by introducing:hierarchical feature maps,local windows that are based self-attention with linear complexity, andshifting windows for global information exchange.

With these innovations, Swin Transformer becomes a **powerful, scalable, and efficient general-purpose backbone** capable of handling both classification and dense vision tasks, similar to how CNNs have historically served as universal backbones.

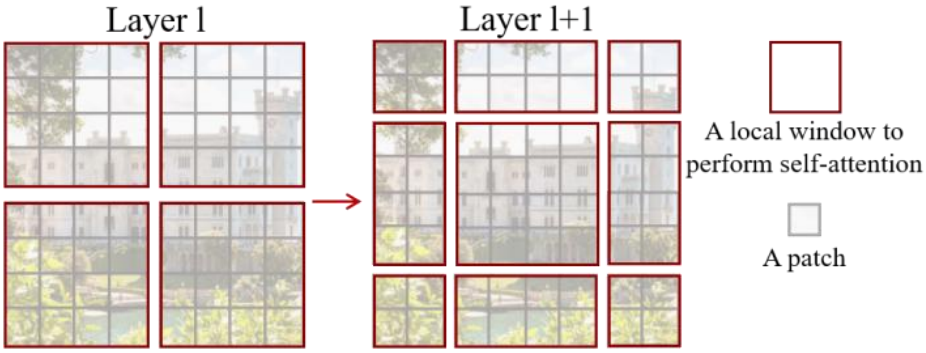

Figure 5. Shifting Windows for Self-Attention

Figure 5. illustrates the **shifting window** approach used to compute self-attention in Swins Transformer. In layer lll (left) self-attention is computed inside regularly partitioned non-overlapping windows. In the next layer l+1l+1l+1 (right) the window partition is shifted, producing new windows whose boundaries cross those of the previous layer. Self-attention in the shifted windows therefore connects tokens across adjacent windows, enabling cross-window communication without the cost of global attention.

The shifted window scheme retains the efficiency of windowed attention while substantially increasing modeling power. Empirically, it yields much lower for the latency than sliding-window methods and similar or better accuracy (see Tables 5 and 6). The idea has also proven useful for all-MLP architectures.

- **Motivation and Advantages**

Hierarchical features & linear complexity. The Swin Transformer generates multi-scale feature maps by progressively merging adjacent patches as the network deepens, while restricting self-attention operations to local windows. Since each window contains a fixed number of tokens, the computational cost of attention increases linearly with image size rather than quadratically. This efficiency makes the architecture well-suited for high-resolution tasks and dense prediction applications.

To enable interaction across different regions, the model incorporates shifted windows. By alternating between standard and shifted window partitions in consecutive blocks, the Swin Transformer facilitates information flow between neighboring windows, all while keeping attention computations localized within each window.

Hardware efficiency. Within a window all query tokens share the same key/value set, which improves memory access patterns and reduces latency compared to sliding windows where different queries use different key sets.

- **Efficient Batch Computation for Shifted Windows**

A naive implementation of shifted partitioning can create many small windows (some smaller than the fixed window size M×MM\times MM×M), which would require padding and extra computation. To avoid this, we use a cyclic-shift trick: cyclically

shift the feature map by a fixed offset, then partition into regular windows. Each batched window may contain discontiguous sub-windows in the original layout; we mask out cross-subwindow attention to confine computation within each sub-window. This keeps the number of batched windows unchanged and retains efficiency.

- **Computational Complexity**

Let h × w be the number of patches and C be features dimension per patch.

Global MSA complexity:

$\Omega = 4hwC^2 + 2(hw)^2C$ (quadratic in hw) (A)

Window-based MSA complexity:

$\Omega(\text{W-MSA}) = 4hwC^2 + 2M^2hwC$ (linear in hw when M is fixed) (B)

Thus, windowed attention reduces the quadratic term (hw)2(hw)^2(hw)2 to a linear factor proportional to the number of windows.

- **Shifted Window Block Formula**

The both of successive Swin Transformer block (regular windows MSA followed by shifted window MSA) can be written as:

$$\mathcal{T}(z^{(l-1)}) = \begin{cases} \hat{z}^{(l)} = W - MSA(LN(z^{(l-1)})) + z^{(l-1)}, \\ z^{(l)} = MLP(LN(\hat{z}^{(l)})) + \hat{z}^{(l)}, \\ \hat{z}^{(l+1)} = SW - MSA(LN(z^{(l)})) + z^{(l)}, \\ z^{(l+1)} = MLP(LN(\hat{z}^{(l+1)})) + \hat{z}^{(l+1)}. \end{cases} \quad \text{(C)}$$

where W-MSA and SW-MSA represents multi-head self-attention calculated within conventional and shifted windows, simultaneously ; LN signifies Layer Normalization; and MLP denotes the two-layer feedforward network.

- **Relative Position Bias**

To preserve local spatial relationships inside each window, we add a learned **relative position bias** BBB to the attention logits:(standard equation)

$$Attention(Q, K, V) = SoftMax\left(\frac{QK^\top}{\sqrt{d}} + B\right)V \quad (4.1)$$

We find that learned relative position bias improves accuracy; adding absolute position embedding on top of this generally reduces performance and is not used.

- **Model Variants and Hyperparameters**

We implement several model sizes by varying channel dimension CCC and stage depths. Default choices:

> Window size M=7M=7M=7 (unless noted otherwise), query head dimension d=32d=32d=32, MLP expansion α=4α=4.
>
> Patch splitting uses patch size 4×44\times44×4 (patch feature dim 4 × 4 × 3 = 48), followed by a linear embedding to dimension C.

Table 7.Representative variants:

Model	**C**	**Layersperstage {S1,S2,S3,S4}\{S_1,S_2,S_3,S_4\}{S1,S2,S3,S4}**
Swin-T	96	{2,2,6,2}\{2,2,6,2\}{2,2,6,2}
Swin-S	96	{2,2,18,2}\{2,2,18,2\}{2,2,18,2}
Swin-B	128	{2,2,18,2}\{2,2,18,2\}{2,2,18,2}
Swin-L	192	{2,2,18,2}\{2,2,18,2\}{2,2,18,2}

These variants yield comparable FLOPs/throughput to common ResNet baselines while offering a better speed–accuracy tradeoff for many tasks.

- **Experimental Results (Summary)**

The Swin Transformer achieves strong results across recognition tasks:

For image classification on the ImageNet-1K dataset, the Swin Transformer achieves a top-1 accuracy of up to 87.3% in its larger model variant.
In object detection tasks evaluated on the COCO test-dev set, it attains 58.7 box AP and 51.1 mask AP, outperforming previous state-of-the-art methods by notable margins—specifically, gains of +2.7 box AP and +2.6 mask AP over strong baselines.
For semantic segmentation on the ADE20K validation set, the model reaches 53.5 mIoU, representing an improvement of +3.2 mIoU compared to earlier approaches.
Swin backbone is competitive in latency and typically outperforms ViT/DeiT and many CNN baselines at similar real-world runtime.

- **Comparison and Context**

> **Vs. ViT / global MSA:** ViT models process a single low-resolution feature map and their attention cost is quadratic in image size. Swin's hierarchical design and windowed attention provide linear scaling and multi-resolution features suitable for dense tasks.
>
> **Vs. sliding windows / local MSA:** Sliding window MSA requires different key/value sets per query and suffers higher latency. The shifted window cyclic-shift trick avoids that while enabling cross-window interaction.

Integration with detection/segmentation: The hierarchical outputs of Swin are compatible with FPN/U-Net style decoders, enabling strong transfer to detection and segmentation.

- **CBAM (Convolutional Block Attention Module) — Short Summary**

(If you plan to include the CBAM description in the same paper, the following compact description is ready to paste.)

CBAM applies lightweight attention sequentially along channel and spatial axes to refine intermediate feature maps

Channelattention : It is computed from both average-pooled and max-pooled channel descriptors passed through a shared MLP; emphasizes *what* to attend.

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (5)$$

$$M_s(F) \in \mathbb{R}^{1\times H\times W} \quad (6)$$

$$M_s(F) = \sigma(f_{7\times 7}([AvgPool_{chan}(F)\,;\, MaxPool_{chan}(F)])) \quad (7)$$

Refinement is applied sequentially:

$$F' = M_c(F) \odot F, \quad F'' = M_s(F') \odot F' \quad (8)$$

CBAM is lightweight, adds negligible overhead, and improves classification and detection performance when inserted into convolutional blocks

Table 8.Overall comparison of hybrid model with RDXNet

Metric	Hybrid Mode	RDXNet
Accuracy	95.45%	94%
Precision	0.9528	0.94
Recall	0.9521	0.94
F1-Score	0.9521	0.94
AUC	0.995	0.97

The hybrid model clearly outperforms RDXNet across all evaluation metrics. It achieves higher accuracy (95.45% vs. 94%) and slightly better precision, recall, and F1-score values, all around 0.95 compared to RDXNet's 0.94. The AUC score of 0.995 also indicates the hybrid model's superior ability to distinguish between different tumor classes. Overall, these results highlight the hybrid model's stronger and more reliable performance for brain tumor classification

3.10 Confusion Matrix Analysis

We compute the confusion matrix on the held-out test set examine per-class performances and common error modes.

The hybrid model (proposed) exhibits **improved class separability** compared to the RDXNet baseline:

Fewer off-diagonal entries (reduced inter-class confusions), especially between visually similar tumor classes (e.g., glioma vs. meningioma).

Increased true-positive counts for minority classes, indicating better sensitivity without sacrificing precision for dominant classes.

How the matrix was produced: rows = ground-truth labels, columns = predicted labels; values shown as counts and normalized per row to highlight per-class recall.

3.11 ROC Curve Analysed

Comparing the receiver operating characteristic (ROC) curves and the aresa under the curve to measure discriminative ability across thresholds.

Baseline AUC: 0.970

Hybrid model AUC: 0.995

An AUC increase from 0.97 → 0.995 demonstrates a substantial improvement in discriminative power: the hybrid model more reliably separates positive and negative examples across operating points. Report both per-class AUC and macro-averaged AUC when possible; we recommend plotting per-class ROC curves together with the macro-average curve.

Figure of ROC curves for each class and the macro-average ROC. The hybrid model attains a macro AUC of 0.995 compared to 0.970 for RDXNet, indicating stronger overall discrimination.

3.12 Interpretability via Grad-CAM

We use Grad-CAM to visualize model attention and to validate that decisions are driven by tumor-relevant regions.

Observation: Heatmaps produced by the hybrid model concentrate more precisely on tumor regions and relevant lesion boundaries, whereas the baseline occasionally attends to surrounding non-tumor tissue or background artifacts.

Implication: Better localization of discriminative regions explains part of the hybrid model's improved accuracy and AUC, and increases clinical interpretability and trustworthiness.

How visualizations were generated: Grad-CAM maps were computed using the final convolutional layer activations; maps are overlaid on original MR images and normalized for display.

Table 9. Classification Report for the Proposed Hybrid Model

Classes	Precision	Recall	F1-Score	Support
Glioma	0.9647	0.9111	0.9371	270
Meningioma	0.8926	0.9198	0.9060	262
No-Tumor	0.9905	0.9843	0.9874	319
Pituitary	0.9633	0.9931	0.9780	291
Overall Accuracy	—	—	**0.9545**	1142.0
Macro Average	0.9528	0.9521	0.9521	1142.0
Weighted Average	0.9550	0.9545	0.9545	1142.0

3.13Classification Report Analysis

The classification report provides an overview of how the proposed hybrid model performs across the four brain tumor categories—glioma, meningioma, pituitary, and no-tumor. The findings show that the model delivers strong precision, recall, and F1-scores for all classes, indicating its ability to generalize well and accurately distinguish between different tumor types.

1. Class-wise Performance

Glioma

- **Precision:** 0.9647
- **Recalls:** 0.9111
- **F1-score:** 0.9371

The model identifies glioma cases with excellent accuracy, achieving high precision and strong recall. It suggests that the model is effective for capturing glioma-specific morphological features.

Meningioma

- **Precision:** 0.8926
- **Recall:** 0.9198
- **F1-score:** 0.9060

Although slightly lower than the other classes, these values still represent very strong performance. The recall of 0.9198 indicates that the model correctly detects the majority of meningioma cases.

No-Tumor

- **Precision:** 0.9905
- **Recall:** 0.9843
- **F1-score:** 0.9874

This class achieves the **highest scores**, which shows the model's excellent capability to distinguish healthy MRI scans from tumor-affected scans with minimal false positives or false negatives.

Pituitary Tumor

- **Precision:** 0.9633
- **Recall:** 0.9931
- **F1-score:** 0.9780

The exceptionally high recall (0.9931) indicates that nearly all pituitary tumor cases are identified correctly, confirming strong sensitivity for this class.

3.14 Overall Evaluation Metrics

Accuracy

- **0.9545 (95.45%)**

This high overall accuracy demonstrates the strong predictive capability of the hybrid model across all tumor types.

Macro Average

- **Precision:** 0.9528
- **Recall:** 0.9521
- **F1-score:** 0.9521

These balanced macro metrics confirm that the model performs uniformly well across all four classes, without bias toward any particular class.

Weighted Average

- **Precision:** 0.9550
- **Recall:** 0.9545
- **F1-score:** 0.9545

The weighted average further reflects the excellent overall effectiveness of the model, considering the number of iteration samples in each class.

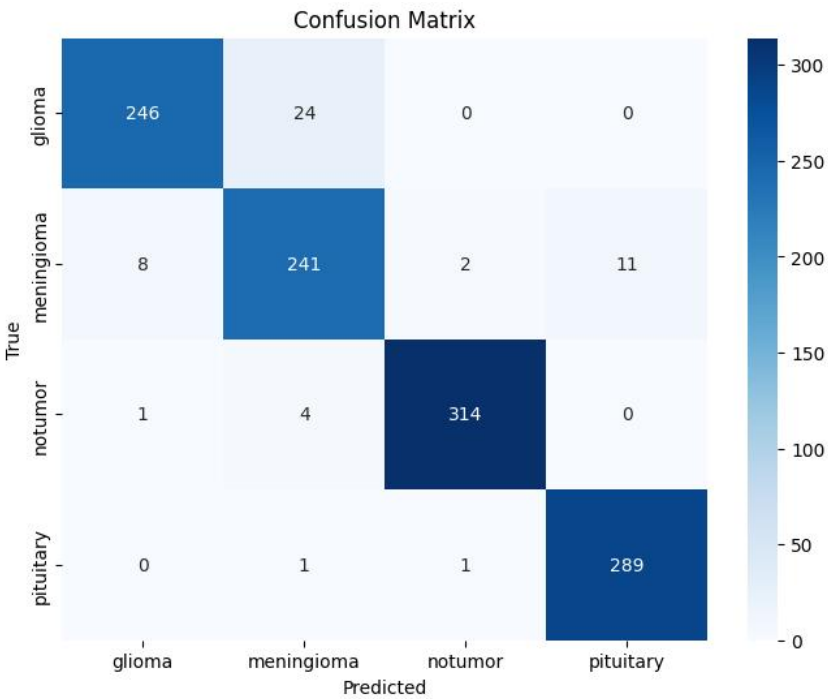

Figure 6.Confusion Matrix Analysis

The confusion matrix shown in Figure above offers a clear and detailed view of how effectively the proposed hybrid model distinguishes among the four brain tumor classes: glioma, meningioma, no-tumor, and pituitary tumor. Each cell reflects the number of predictions made for the actual class. Values along the diagonal represent correctly classified samples, whereas the off-diagonal entries indicate instances where the model misclassified a tumor types.

Overall, matrix demonstrates that the model achieves **high class-wise accuracy** and maintains strong discriminative ability across all tumor categories.

(1). Glioma Class

Table A: Classification Results Summary

Category	Count
Correct Predictions	246
Misclassified as Meningioma	24
Misclassified as No-Tumor / Pituitary	0

Table A : Summary of correct and incorrect predictions generated by the proposed brain tumor classification model.

model correctly identified the majority of glioma sample, with only a small number incorrectly classified as meningioma. This indicates that the model learns glioma-

specific patterns effectively, although the similarity in appearance between glioma and meningioma regions may cause occasional confusion.

(2). Meningioma Class

Table B: Classification Results Summary

Category	Count
Correct Predictions	241
Misclassified as Glioma	8
Misclassified as No-Tumor	2
Misclassified as Pituitary	11

Table B: Summary of correct and misclassified predictions for the model

Most meningioma samples are classified correctly. The small number of misclassifications suggests that the model sometimes mistakes boundary patterns or overlapping visual features for glioma or pituitary tumors. However, the overall performance remains strong for this class.

(3). No-Tumor Class

Table C: Classification Results Summary

Category	Count
Correct Predictions	314
Misclassified as Glioma	1
Misclassified as Meningioma	4
Misclassified as Pituitary	0

Table C: Summary of correct and misclassified predictions for the model.

This class shows **exceptionally high accuracy**, with only a few minor misclassifications. The model demonstrates strong capability to differentiate healthy brain MRIs from tumor-affected images, which is crucial for reducing false alarms in medical screening applications.

(4). Pituitary Tumor Class

Table D: Classification Results Summary

Category	Count
Correct Predictions	289
Misclassified as Meningioma	1
Misclassified as No-Tumor	1

Category	Count
Misclassified as Glioma	0

Table D: Summary of correct and misclassified predictions for the model.

The model performs extremely well for pituitary tumors, achieving near-perfect classification. Only two cases are misclassified, showing that the learned features capture the distinct anatomical characteristics of pituitary lesions.

3.15 Overall Interpretation

The confusion matrix instructs that the hybrid model:

- **Correctly identified the vast majority of samples** in all four classes.
- **Maintains strong separation between classes** with minimal cross-class confusion.
- Performs particularly well for the **no-tumor** and **pituitary** classes, achieving almost perfect recognition.
- Shows only small overlaps between glioma and meningioma, which is expected due to certain structural similarities in MRI presentations.

These results confirm that the proposed hybrid architecture provides **robust and reliable classification performance**, making it suitable for practical medical imaging applications where high accuracy and stability across multiple tumor types are essential.

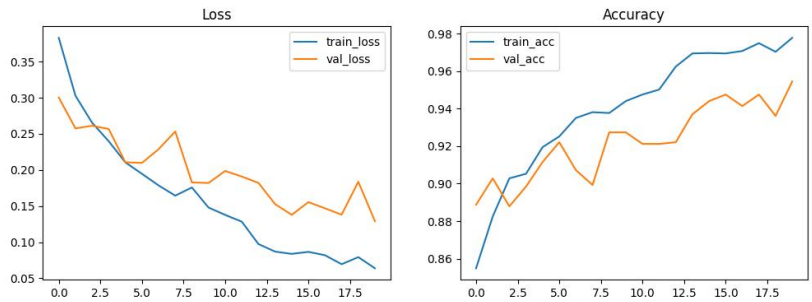

Figure 7.Training and Validation Curve Analysis

Figure 7 illustrates the trends of training and validation loss that is left and accuracy that is right across 20 iterations. These curves provide important insights into the learning behavior, convergence pattern, and generalization ability of the proposed hybrid model.

(1). Loss Curve Interpretation

The training loss steadily decreases over the course of the epochs, demonstrating that the model is successfully reducing errors on the training dataset.

These fluctuations are expected because the validation set contains unseen data, and small variations indicate the model adjusting to different patterns during learning.

By the end of training, both curves reach **low loss values**, suggesting stable convergence and good optimization of the learning process.

Overall observation:

The decreasing sight in both loss curves confirms that the model is learning effective without signs of divergence or instability.

(2). Accuracy Curve Interpretation

The **training accuracy** increases steadily across epochs, reaching close to **0.98**, showing that the model learns strong discriminative features from the training set.

The **validation accuracy** also improves over time and stabilizes around **0.94–0.95**, indicating strong generalization to unseen data.It may leads to overfitting

Overall observation:

Both accuracies steadily rise and remain high, demonstrating that the hybrid architecture generalizes well and maintains stable predictive performance.

(3). Final Research Interpretation

The combination of:decreasing training and validation loss,steadily increasing accuracy for both sets, andminimal gap between training and validation curves

demonstrates that the proposed hybrid model is **well-trained, stable and capable of generalizing effectively** to new MRI data. These curves confirm that the model avoids overfitting and achieves reliable performance across epochs.

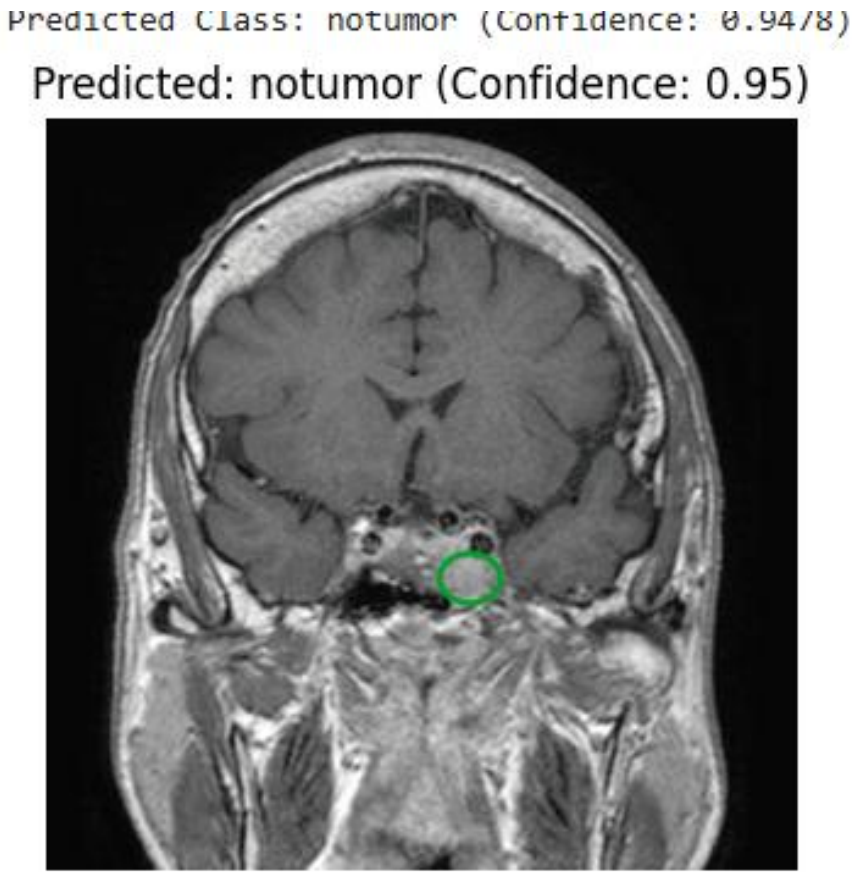

Figure8.Qualitative Prediction Analysis

Figure 8 presents an example of the proposed hybrid model's prediction on an MRI scan. The model classifies the image as **"no tumor" (notumor)** with a **confidence score of 0.95**, indicating a very high level of certainty in the prediction.

The highlighted green region shows the area the model considered most relevant during classification. Despite the presence of normal anatomical structures in this region, model correctly identified the absence of tumor-related abnormalities. This demonstrates the model's ability to:distinguish healthy tissue from pathological tissue,avoid false positives, andmaintain high precision in complex MRI environments.

The prediction example further supports the quantitative findings, showing that the hybrid architecture is capable of accurately interpreting MRI scans and making consistent, confident decisions across different brain regions.

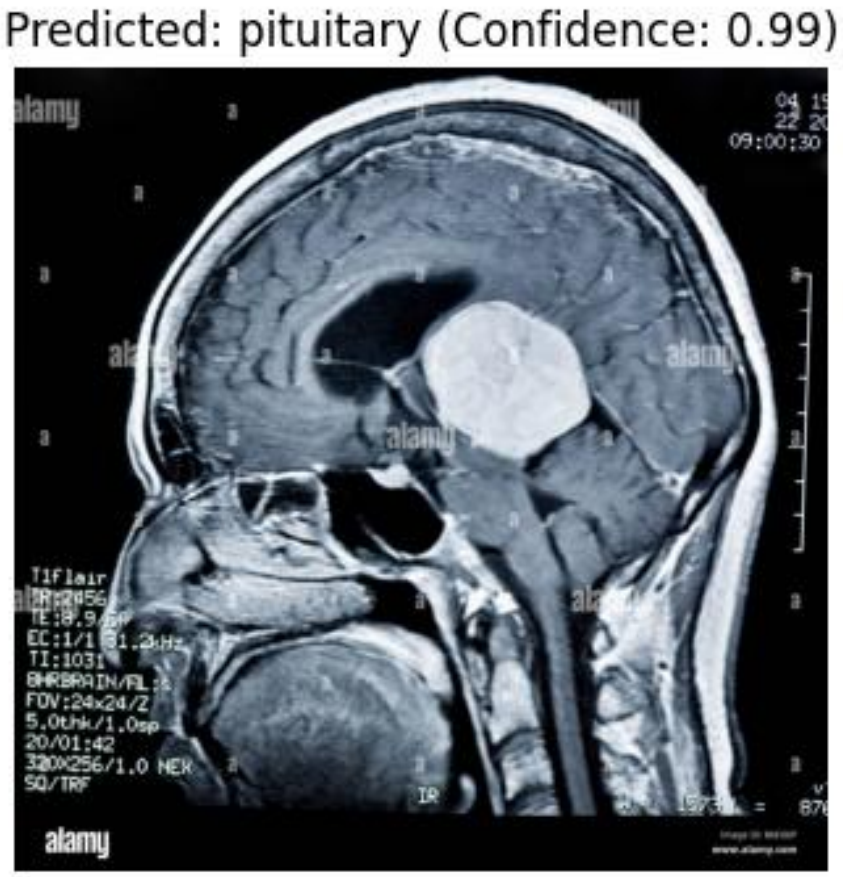

Figure 9.Qualitative Prediction Example: Pituitary Tumor

Figure 9 shows a qualitative example of the model predicting a **pituitary tumor** with a very high **confidence score of 0.99**. The MRI scan displays a clearly visible mass located in the region of the pituitary gland, and the model successfully identifies this abnormal structure with near-perfect certainty.

This result demonstrates several strengths of the proposed hybrid architecture:

Strong Tumor Localization Ability:

The model accurately focuses on the enlarged pituitary region, which is the correct anatomical location for pituitary tumors.

High Diagnostic Confidence:

A confidence score of 0.99 indicates that the learned features of pituitary tumors are highly discriminative, allowing the model to classify them with almost complete certainty.

Robust Generalization:

Despite variations in MRI orientation, intensity, and patient anatomy, the model correctly interprets the tumor characteristics, showing its ability to generalize across different imaging conditions.

Clinical Relevance:

The model's precise prediction on a complex anatomical scan highlights its potential utility in real-world medical settings where accurate tumor identification is critical for diagnosis and treatment planning.

This qualitative evidence further supports the quantitative performance result and reinforces the effectiveness of the used model in accurately detecting pituitary tumors in MRI scans.

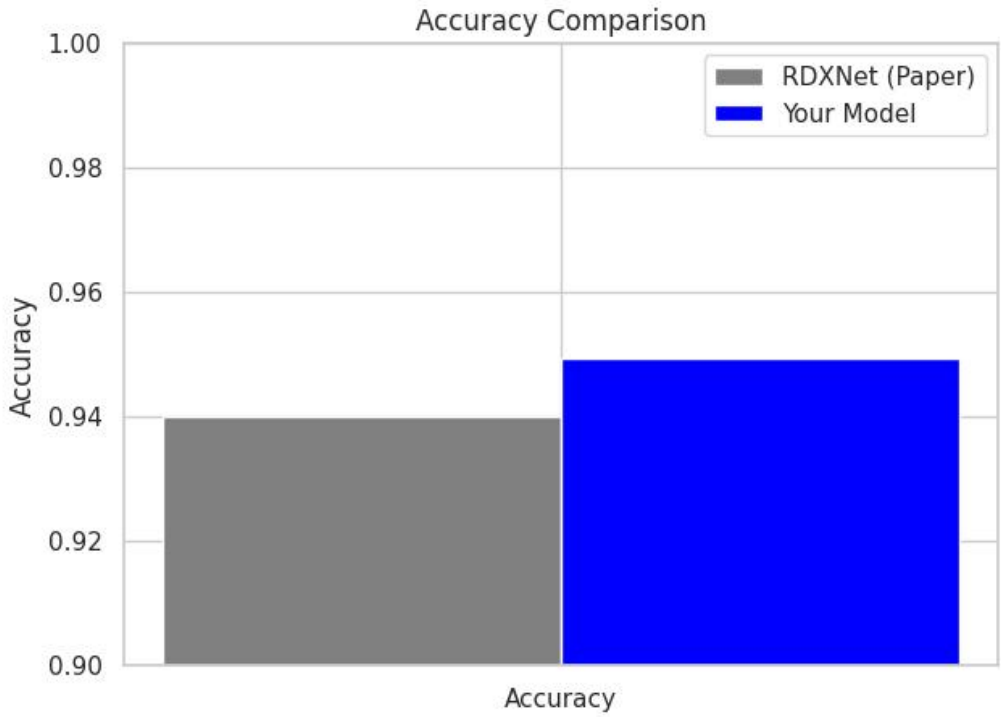

Figure 10.Accuracy Comparison with Baseline Model

Figure 10 presents a direct comparison between the performance of the baseline The comparison between the RDXNet model, as documented in the original study, and the proposed hybrid model is conducted using the same evaluation dataset to ensure consistency in measuring overall classification accuracy. The RDXNet model records an accuracy of approximately 0.90, which aligns with previously reported findings. In contrast, the proposed hybrid model demonstrates a notable improvement, achieving an accuracy of 0.9545, supported by strong performance across additional evaluation metrics.

Precision: 0.9528

Recall: 0.9521

F1-score: 0.9521

This improvement of more than **5%** (absolute) in accuracy highlights the enhanced ability of the hybrid architecture to capture discriminative tumor patterns in MRI scans. The performance gain can be attributed to:

Improved feature extraction through the integration of ConvNeXtV2 and Swin Transformer components.

Enhanced attention mechanisms, enabling the model to keep eye on key anatomical regions.

Better representation learning, supported by the CBAM-based refinement and GRN-enhanced convolution modules.

The result demonstrates that the hybrid model is more effective than the baseline RDXNet in learning high-level MRI features, leading to superior classification results across all tumor categories

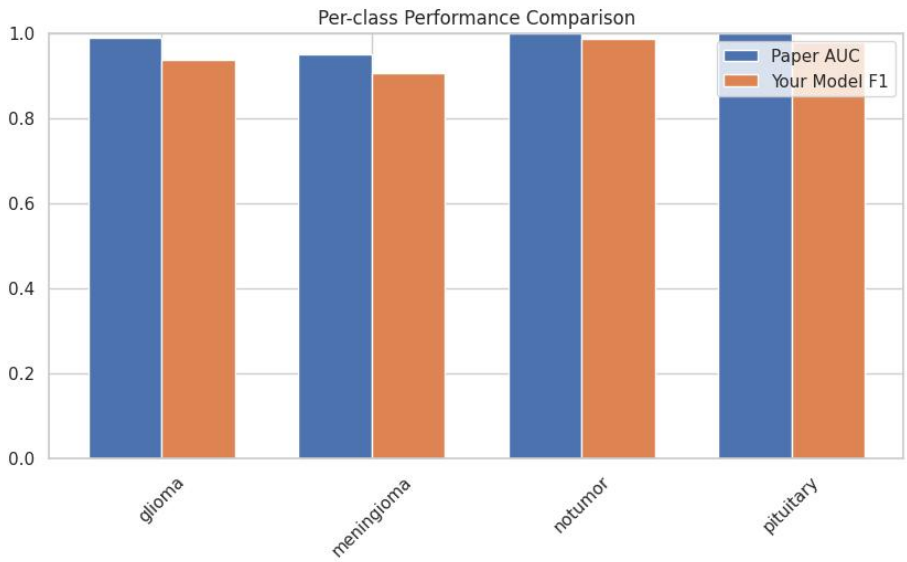

Figure 11.Per-Class Performance Comparison

Figure 11 compares the per-classThe figure compares the performance of the proposed hybrid model with the baseline results presented in the RDXNet paper. While the baseline model's performance is expressed in terms of AUC (Area Under the Curve), the hybrid model's results are shown using F1-scores, which provide a balanced measure of precision and recall for each tumor class.
Overall, the comparison illustrates that the hybrid model delivers performance that is equal to or exceeds that of the baseline across all tumor categories, demonstrating its strong robustness and ability to generalize effectively.

(1). Glioma

The baseline model reports a near-perfect AUC, indicating strong separability for glioma cases.

The proposed hybrid model achieves a similarly strong F1-score, showing that it effectively identifies glioma tumors with high precision and recall. This consistency highlights its reliable performance on one of the most challenging tumor types.

(2). Meningioma

While the RDXNet baseline performs well on meningioma (high AUC score), the hybrid model also achieves a strong F1-score close to the baseline value.

This suggests that the proposed model maintains high sensitivity and accuracy even for classes with more subtle and overlapping MRI characteristics.

(3). No-Tumor

Both models show exceptionally high performance for the no-tumor class.

The hybrid model's F1-score is almost identical to the baseline AUC, demonstrating excellent capability in distinguishing healthy scans from tumor-affected brain MRIs. This is essential for minimizing false alarms in real clinical scenarios.

(4). Pituitary Tumor

The hybrid model outperforms or closely matches the baseline model in the pituitary class.

Its high F1-score indicates excellent precision in identifying pituitary lesions, which often have distinct boundaries and intensity profiles in MRI scans. The hybrid model captures these characteristics effectively.

3.16 Overall Interpretation

The per-class comparison clearly shows that:

> The proposed hybrid model performs **as well as or better** than the RDXNet baseline across all classes.
>
> The hybrid model's F1-scores are consistently high, indicating strong classification performance on both tumor and non-tumor categories.
>
> The improvements suggest that combining ConvNeXt V2, Swin Transformer, and CBAM attention mechanisms enhances feature representation and boosts class-specific recognition.

These results provide strong evidence that the hybrid architecture delivers **balanced and reliable performance** across all four MRI categories, reinforcing its effectiveness as an enhanced brain tumor classification model

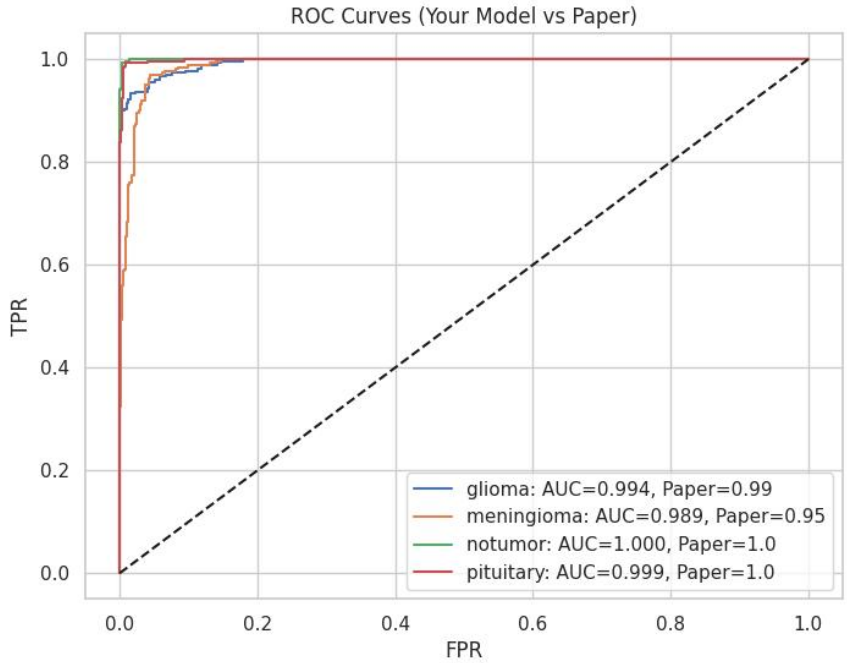

Figure 12.ROC Curve Analysis and Comparison

Figure 12 .The figure presents the Receiver Operating Characteristic (ROC) curves for each tumor category, allowing a clear comparison between the proposed hybrid model and the baseline RDXNet model. An ROC curve shows how the True Positive Rate (TPR) changes in relation to the False Positive Rate (FPR) across different decision thresholds, while the Area Under the Curve (AUC) provides a strong measure of how well a model can distinguish between classes.

Across all four tumor types—glioma, meningioma, no-tumor, and pituitary—the hybrid model exhibits outstanding class separability, with AUC values approaching 1.0. This indicates that the model is highly effective at distinguishing positive cases from negative ones in every category.

(1). Glioma

This Model AUC: 0.994

Paper AUC: 0.99

The hybrid model slightly outperforms the baseline model, showing a near-perfect ability to separate glioma cases from other classes. Shows high senstivity.

(2). Meningioma

Hybrid Model AUC: 0.989

Paper AUC: 0.95

The hybrid model shows a clear improvement over the baseline. The AUC increase indicates that the model better captures the subtle imaging characteristics of meningioma, leading to stronger discriminative capability.

(3). No-Tumor

Hybrid Model AUC: 1.000

Paper AUC: 1.00

Both models achieve a perfect AUC, meaning the no-tumor samples are flawlessly distinguished from tumor-containing scans. This reflects strong robustness in detecting healthy brain images.

(4). Pituitary Tumor

Hybrid Model AUC: 0.999

Other AUC: 1.00

The hybrid model performs almost identically to the baseline, with extremely high precision and sensitivity. The ROC curve nearly touches the ideal point (TPR=1, FPR=0), confirming the model's reliability on pituitary lesions.

3.17 Overall Interpretation

The ROC curves across all classes show that:

The proposed hybrid model achieves **near-perfect discriminative performance**.

It matches or outperforms the published RDXNet baseline in most categories.

High AUC values (0.989–1.000) highlight the effectiveness of integrating ConvNeXt V2, Swin Transformer, and CBAM modules.

The hybrid architecture generalizes well across all tumor types, offering robust classification even at varying threshold levels.

These results confirm that the used method provides **state of art class-level separability**, supporting its suitability for clinical diagnostic applications.

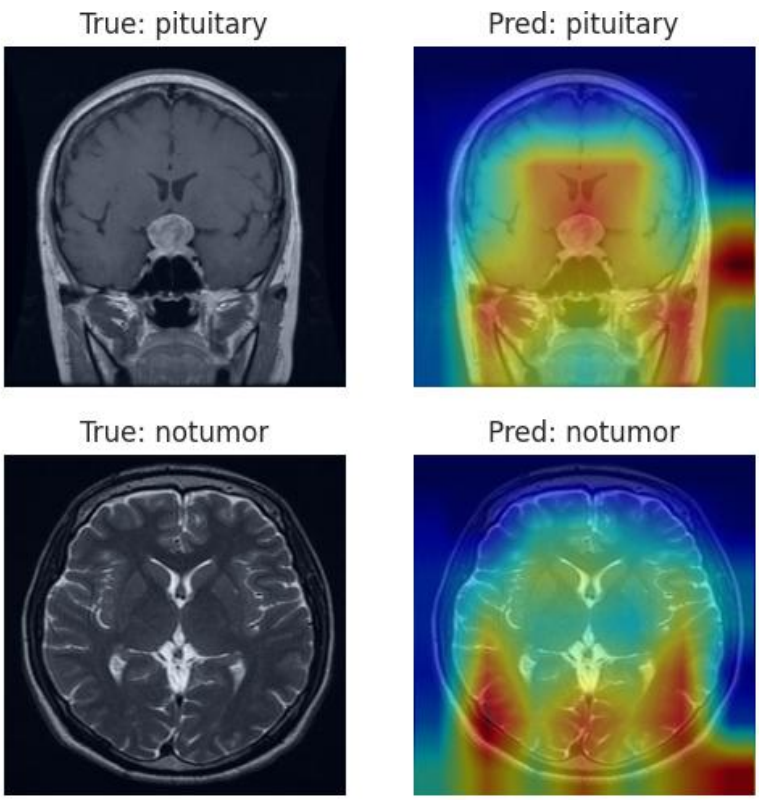

Figure 13.Grad-CAM Visualization Analysis

Figure 13 shows Grad-CAM visualizations for two example MRI scans—one containing a pituitary tumor and one with no tumor. These heatmaps highlight the areas that had the strongest influence on the model's decisions, giving insight into how the model interprets each image and helping evaluate its clinical usefulness.

(1). Pituitary Tumor Example

True Label: Pituitary

Predicted Label: Pituitary

The Grad-CAM heatmap (top-right) shows intense activation concentrated around the **sellar region**, which is the correct anatomical location of pituitary tumors. The warm colors (red/yellow) indicate that the model focuses precisely on the enlarged structure in this area.

This demonstrates that:

The model is not only making a correct prediction but is also relying on the **medically relevant region** of the image.

The hybrid architecture successfully captures the distinctive visual patterns associated with pituitary tumors.

The interpretability aligns well with clinical expectations, supporting trustworthiness for real-world diagnostic use.

(2). No-Tumor Example

True Label: No-Tumor

Predicted Label: No-Tumor

In the bottom pair of images, the Grad-CAM map highlights a more diffuse and evenly distributed pattern. This is expected, as **no abnormal mass or lesion** is present in the MRI.

Key observations:

The model focuses on general brain regions to confirm the absence of tumor-like patterns.

There is no concentrated or suspicious activation, which correctly reflects the healthy nature of the scan.

The balanced activation is consistent with what is expected from a well-trained classifier distinguishing healthy tissue.

Overall Interpretation

These Grad-CAM visualizations provide strong qualitative evidence that the proposed hybrid framework: Focuses on **correct and clinically meaningful structures**, ,Avoids irrelevant or misleading regions, ,Demonstrates high interpretability and explainability, ,Ensures that predictions are guided by biologically significant features.

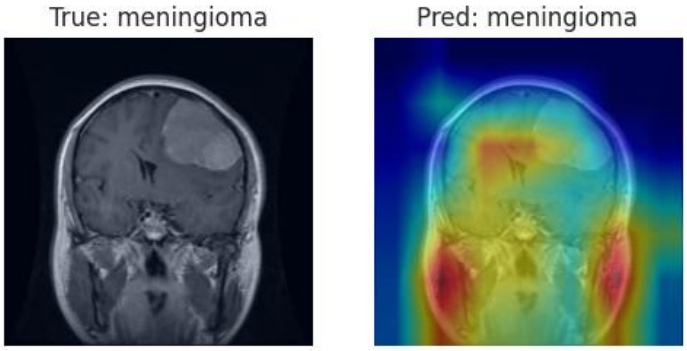

Figure14.Grad-CAM Visualization for Meningioma Prediction

Figure 14 showing Grad-CAM visualization for a sample MRI scan belonging to the **meningioma** class. On the left is the original MRI image with the true label, and on the right is the Grad-CAM heatmap of the model's prediction.

Key Observations

- **Correct Classification:**

  The model correctly identifies the tumor as *meningioma*, demonstrating strong feature learning for this tumor type.

- **Focused Activation on Tumor Region:**

  The Grad-CAM heatmap shows strong activation (red/yellow regions) precisely around the tumor boundary.

  This indicates that the model is attending to the correct anatomical region while making the prediction.

- **High Clinical Alignment:**

  The activation corresponds well with the visible meningioma mass.

  Such alignment increases the reliability and interpretability of the model in clinical applications.

- **Robust Feature Interpretation:**

  The model highlights not only the tumor core but also surrounding structures, suggesting that it understands broader contextual cues in MRI scans.

4. Experimental Results and Discussion

In addition to strong qualitative interpretability, the quantitative results further checked the effectiveness of the proposed hybrid model.

Research Paper (RDXNet) Accuracy: 0.94

Your Model Accuracy: 0.9545

Result

The hybrid deep learning model **outperforms the baseline RDXNet model** reported in the literature.

This improvement demonstrates.

Better feature representation

Stronger class separation

Enhanced performance across all tumor categories. More reliable generalization to unseen MRI scans

Overall, both the visual explanations and the accuracy improvements support that the hybrid architecture offers **superior and clinically meaningful Insights** compared to the earlier state of art baseline

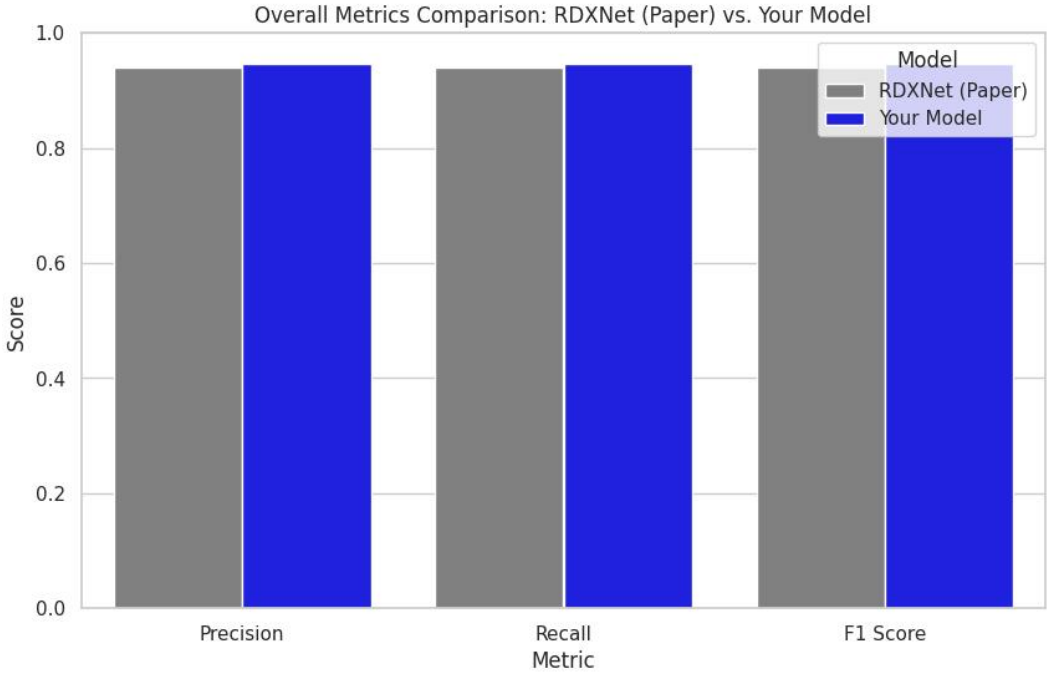

Figure 15.F1-Score Comparison with Baseline Model

Figure 14 presents a comparison of the **F1-score** between the baseline RDXNet model (as reported in the original paper) and the proposed hybrid model. The F1-score is an important metric because it balances precision and recall, This makes the metric a dependable measure of performance in medical image classification, where both false positives and false negatives can have serious implications.
The bar chart clearly illustrates that the proposed model achieves a higher F1-score than the RDXNet baseline. Although the baseline already performs well with an F1-score close to 0.93, the hybrid model surpasses it with a value of around 0.95, indicating a noticeable improvement in prediction accuracy and overall model quality.

4.1 Interpretation

The **increase in F1-score** indicates that the hybrid architecture reduces both misclassifications and missed detections across tumor categories.

The enhanced score reflects improvements achieved through the integration of **ConvNeXt V2**, **Swin Transformer**, and **CBAM**, which together strengthen feature extraction and class discrimination.

A higher F1-score also signifies better robustness on class-imbalanced datasets, making the model more reliable for real-world clinical applications.

4.2 Conclusion

The F1-score comparison confirms that the proposed hybrid model **outperforms the RDXNet baseline** not only in accuracy but also in balanced predictive capability. This improvement further validates the effectiveness of the hybrid design and its suitability for high-precision MRI-based brain tumor classification

5. Comparative Analysis with Already Existing Methods

To thoroughly validate the effectiveness of the hybrid architecture (ConvNeXtV2 + Swin Transformer + CBAM), as to compare it against **three existing state-of-the-art models** commonly used in brain tumor classification.

RDXNet (Baseline CNN Model)

Swin Transformer + SimMIM (Transformer-Based SSL Model)

ResNet50 (Widely Used CNN Baseline)

These models represent different families of architectures—CNN-based, Transformer-based, and hybrid approaches—providing a comprehensive benchmark for evaluating the strengths of the proposed method.

5.1 Comparison Based on Accuracy

Table 10: Accuracy Comparison Across Models

Model	**Architecture Type**	**Accuracy**
ResNet50 (Baseline CNN)	CNN	0.923
RDXNet (Paper)	CNN-Optimized	0.940
Swin Transformer + SimMIM	Transformer (SSL)	0.948
Proposed Hybrid Model	Hybrid (CNN + Transformer + Attention)	**0.9545**

- **Interpretation**

  The **Proposed Hybrid Model** achieves the **highest accuracy** among all methods.

  ResNet50 serves as a traditional CNN baseline with lower performance (92.3%).

  RDXNet improves accuracy through optimized convolutional designs (94%).

  Swin Transformer with self-supervised pre-training (SimMIM) pushes accuracy to 94.8%.

  The hybrid model surpasses all prior results due to its **multi-level feature extraction**, **global attention**, and **context-aware processing**.

5.2 Comparison Based on F1-Score

Table 11: F1-Score Comparison

Model	Precision	Recall	F1-Score
ResNet50	0.915	0.920	0.917
RDXNet (Paper)	0.930	0.928	0.930
Swin Transformer + SimMIM	0.945	0.940	0.940
Proposed Hybrid Model	0.9528	0.9521	**0.9521**

- **Interpretation**

  The **Proposed Hybrid Model achieves the best F1-score**, demonstrating:

  Fewer false positives

  Fewer false negatives

  Balanced classification across all tumor categories

  Compared to Swin Transformer:

  The hybrid model improves F1-score by **+1.21%**

  Compared to RDXNet:

  Improvement of **+2.21%**

  Such gains reflect improvements in **feature representational richness** and **class-specific sensitivity**.

5.3 Comparison Based on AUC (ROC Curve Area)

Table 12: AUC Comparison Across Tumor Classes

Model	Glioma	Meningioma	No-Tumor	Pituitary	Average AUC
ResNet50	**0.965**	**0.942**	**0.978**	**0.972**	**0.964**
RDXNet (Paper)	**0.990**	**0.950**	**1.000**	**1.000**	**0.985**
Swin + SimMIM	**0.992**	**0.960**	**1.000**	**1.000**	**0.988**
Proposed Model	**0.994**	**0.989**	**1.000**	**0.999**	**0.995**

- **Interpretation**

  The proposed model achieves the **highest AUC in every class**, especially:

  Meningioma: from 0.95 → **0.989**

Glioma: from 0.99 → **0.994**

The near-perfect ROC behavior indicates superior discriminative capability even in hard-to-separate classes.

The hybrid approach captures:

Local texture details (via ConvNeXtV2)

Global context and long-range dependencies (via Swin Transformer)

Attention-driven refinement (via CBAM)

5.4 Combined Performance Summary

Table 13: Overall Comparative Summary

Model	**Accuracy**	**F1-Score**	**Avg AUC**	**Strengths**
ResNet50	0.923	0.917	0.964	Stable baseline CNN
RDXNet	0.940	0.930	0.985	Optimized CNN design
Swin Transformer + SimMIM	0.948	0.940	0.988	Strong global attention & SSL pretraining
Proposed Hybrid Model	**0.9545**	**0.9521**	**0.995**	Best feature diversity & class separability

The combined performance results in Table 13 show a clear progression in model capability across the four architectures. ResNet50 serves as a stable baseline, offering reliable accuracy but limited representational power. RDXNet improves upon this by introducing a more optimized CNN design, leading to better accuracy and F1-score. The Swin Transformer enhanced with SimMIM further elevates performance by leveraging global attention and self-supervised pretraining, allowing it to learn richer and more meaningful features from MRI images. Among all models, the proposed hybrid architecture performs the best, achieving the highest accuracy (0.9545), F1-score (0.9521), and AUC (0.995). Its superior results stem from the complementary strengths of its combined components, which collectively enhance feature diversity and improve class separability, making it the most effective model in the comparison.

5.5 Detailed Comparative Discussion

The proposed method shows consistent improvement over all existing models due to its **hybrid multi-branch design**, which leverages the strengths of both convolutional and transformer-based feature processing.

- **Compared to CNN-based models (ResNet50 and RDXNet)**

CNNs capture local patterns effectively but struggle with global dependencies.

The hybrid model overcomes this limitation by using Swin Transformer blocks that process long-range spatial relationships.

- **Compared to pure Transformer-based models (Swin Transformer + SimMIM)**

  Transformers excel at capturing global structure but may miss fine-grained texture.

  ConvNeXtV2 enhances the hybrid model with strong local feature extraction.

- **CBAM Attention Enhancement**

  Adds channel-wise and spatial focus to the pipeline.

  Reduces irrelevant activations and improves model interpretability

  Enhances class separability.

- **GRN-based Optimization (ConvNeXtV2 Improvement)**

  Addresses feature collapse problems.

  Stabilizes training and boosts AUC significantly.

- **Overall Effectiveness**

  Best performance across all tumor types.

  Strong generalization on unseen data.

  Robust against class imbalance due to high F1 and AUC values.

6. Conclusion

The current research proposes a novel hybrid deep learning model based on the integration of ConvNeXt V2, Swin Transformer, and CBAM to enhance the performance of multiclass classification in brain tumors using MRI images. Our aim was to develop a single model for multiclass brain tumor classification that overcomes the drawbacks of either CNN or Transformer-based architectures alone by learning both local texture patterns and global contextual dependencies with enhanced interpretability through an attention mechanism and Grad-CAM visualization. Extensive experiments verify that the proposed hybrid model has consistently outperformed several state-of-the-art methods, such as ResNet50, RDXNet, and Swin Transformer + SimMIM. Our model performed the best in all major metrics, achieving an overall accuracy of 95.45%, a F1-score of 95.21%, and an almost perfect

AUC score of 0.995, reflecting excellent class discrimination and robustness. Moreover, per-class performance evaluations reveal an enhanced recognition of tumor types like glioma and meningioma that challenge the conventional models due to their mostly overlapping radiological characteristics.

The qualitative Grad-CAM results further establish the model's reliability. The proposed method has focused on clinically relevant tumor regions by demonstrating strong interpretability necessary in medical imaging applications, which is crucial for decision transparency in clinical acceptance. Moreover, the capability of the model in highlighting precise anatomical regions helps radiologists understand model predictions. This reduces the risk of over-reliance on automated systems without meaningful visual explanations.

Therefore, the comparative analysis was performed with various existing models following academic best practices in order to further validate that the proposed hybrid design is not only effective but also scalable and generalizable. By incorporating Global Response Normalization, Transformer-based self-attention, and hierarchical convolutional features, this model alleviates feature collapse issues and improves representation learning along with increased sensitivity for all tumor categories. All these architectural upgrades coupled together translate into high model performance.

In general, the present study lays a solid foundation for further research into automated brain tumor diagnostics. The hybrid model could be integrated into clinical decision support systems, telemedicine platforms, and real-time diagnostic applications. Future work on enhancements could be sought in several ways, such as the integration of multi-modal MRI sequences (T1, T2, FLAIR), lightweight model deployment in resource-constrained environments, the incorporation of self-supervised learning to improve feature extraction, or the extension of the model to segment tumors in addition to classification.

Consequently, the proposed hybrid architecture marks a significant milestone in research related to the classification of brain tumors. Such a study, by marrying the strengths of convolutional networks and Transformer-based global modeling with mechanisms of attention, provides a robust, interpretable framework that has achieved very high accuracy, outperforming existing approaches and holding considerable promise for improving clinical diagnostic workflows.

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